

TECHNICAL TRAINING PROGRAM

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Training Days 2026 –
Windows OS

CHAPTER 2 – WINDOWS OPERATING SYSTEM

INTRODUCTION TO WINDOWS OS

Microsoft Windows is one of the most widely used Desktop Operating Systems in homes, schools, and business environments. It provides the working environment through which users interact with applications, files, devices, and system resources.

Before graphical versions of Windows became common, many personal computers used **Microsoft Disk Operating System**, better known as **MS-DOS**. MS-DOS was mainly operated through a Command-Line Interface. Users entered text commands to start programs, manage files, and perform system tasks.

The drive letters still used in Windows have historical roots. The letters **A:** and **B:** were traditionally assigned to floppy disk drives. When hard disks became common in personal computers, the primary hard drive was assigned the letter **C:** This naming convention continues to be used today, even though floppy disk drives are no longer common.

Windows introduced a graphical working environment that made computers easier for many users to understand and operate. Instead of relying only on typed commands, users could interact through windows, icons, menus, folders, and a pointer. This type of environment is known as a **Graphical User Interface (GUI)**.

Windows is a proprietary Operating System developed and distributed by Microsoft. Modern versions of Windows are based on the **Windows NT** architecture and contain several important system components.

The kernel is the central part of the Operating System. It manages essential resources such as processor time, memory, hardware devices, and input/output operations. It also helps applications communicate safely with the computer's hardware.

The user interface is the part of Windows that users see and interact with. It includes the desktop, Start menu, taskbar, windows, icons, files, folders, Settings, and other graphical tools.

Although most users work through the graphical interface, Windows also provides command-line tools such as *Command Prompt* and *PowerShell*. These tools are useful for administration, troubleshooting, configuration, and automation.



WINDOWS PROS & CONS

Pros

- **Wide Adoption**: Windows is widely used in homes, schools, and business environments.
- **Broad Hardware Support**: It supports a large range of computers, peripheral devices, and hardware components.
- **Extensive Software Compatibility**: Many business, productivity, communication, creative, and gaming applications are developed for Windows.
- **Familiar User Interface**: Many users are already familiar with the Windows desktop, Start menu, taskbar, and File Explorer.
- **Enterprise Management Tools**: Windows supports centralized administration, user management, security policies, updates, and integration with business environments.
- **Vendor and Community Support**: Technical documentation, professional support, training resources, and user communities are widely available.

Cons

- **Licensing Costs**: Windows editions and some related business tools may require paid licences.
- **Hardware Requirements**: Newer Windows versions may not support older hardware or may require more processor, memory, and storage resources.
- **Security Target**: Because Windows is widely used, it is frequently targeted by malware and attackers. Security depends on correct configuration, updates, access control, and user awareness.
- **Update Management**: Some updates may require restarts, cause temporary compatibility issues, or need testing before deployment in a business environment.
- **Limited Source-Code Access**: Windows is proprietary software, so users and organizations cannot freely inspect or modify its source code.
- **Compatibility Differences**: Older applications, drivers, or specialist devices may not work correctly with every Windows version.

Think about this:

Why might an organization choose Windows even when another Operating System has lower licensing costs?

BASIC WINDOWS NAVIGATION

In this section, you will become familiar with the main areas of the Windows interface, including the **desktop**, **taskbar**, **Start menu**, **File Explorer**, **Settings**, and **Control Panel**.

You will practice opening and managing applications, creating and organizing folders, moving and renaming files, using search, switching between windows, and applying common keyboard shortcuts.

The aim is not only to recognize the Windows interface, but also to use it efficiently during everyday technical and administrative tasks.

DESKTOP, TASKBAR & START MENU – A QUICK GUIDE

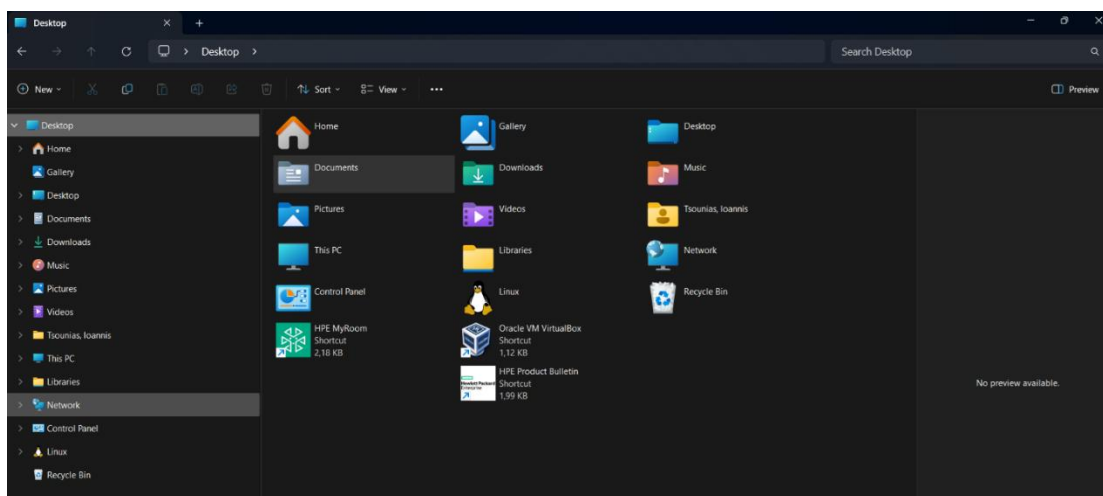
Quick navigation overview: Desktop, Taskbar, and Start menu

FILE & FOLDER MANAGEMENT

WINDOWS FILE EXPLORER (WIN + E)

File Explorer is the main Windows tool used to *view*, *organize*, *search*, *copy*, *move*, *rename*, and *delete* files and folders.

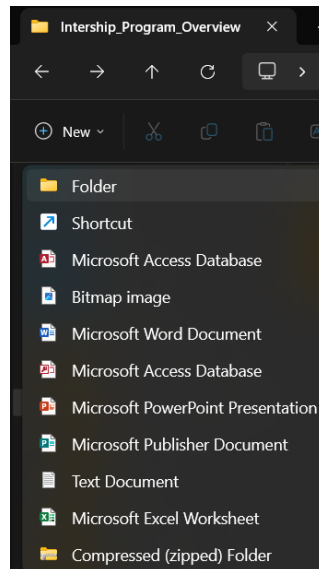
It presents storage locations in a hierarchical structure, allowing users to navigate between drives, folders, and subfolders.



Windows File Explorer Folders, Subfolders, Preview pane



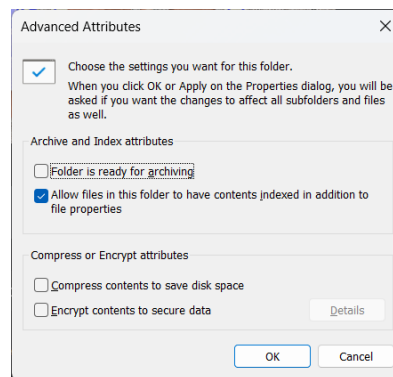
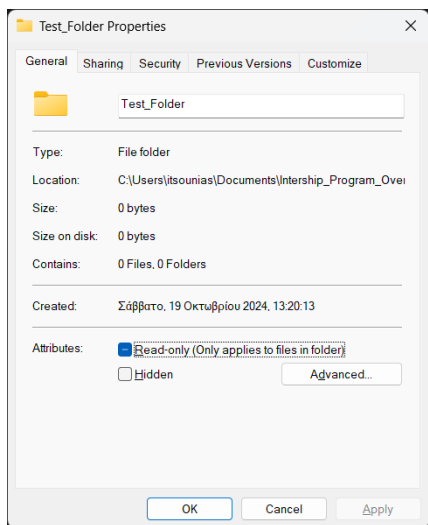
CREATE A NEW FOLDER IN FILE EXPLORER



Create a folder: Select New > Folder, or Right-click inside File Explorer and select New > Folder

“Give the folder a clear and meaningful name. Avoid vague names such as “New Folder” or “Files” when the purpose of the folder is known.”

TAB: GENERAL



Right Click > Properties > Advanced

ADVANCED ATTRIBUTES EXPLAINED

Folder is “Ready for Archiving”: This attribute indicates that the folder or file has changed since it was last marked as backed up. Some backup applications can use this information to decide which data should be included in the next backup operation.

Indexing: When indexing is enabled, Windows Search may include the file names, properties, and—in supported file types—the contents of files. This can make searches faster and more accurate.

Compress contents to save disk space: Windows can use **NTFS compression** to reduce the amount of disk space used by files and folders. The data is compressed and decompressed automatically when it is accessed.

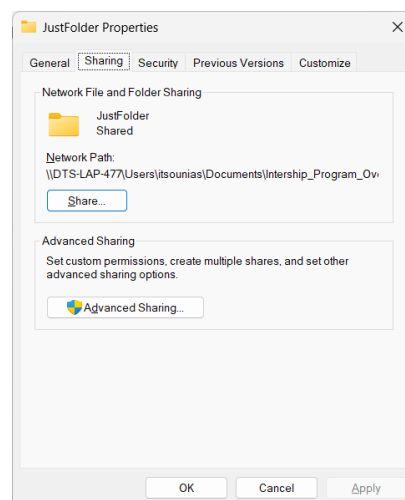
Compression may save storage space, but it can also require additional processor work and may not provide significant savings for files that are already compressed, such as many images, videos, and archive files.

Encrypt contents to secure data: This option uses the **Encrypting File System (EFS)** to protect supported files and folders. The encrypted data is normally accessible only to the user account that encrypted it or to an authorized recovery account.

Encryption protects the contents of the data, but it does not replace backups. If the encryption certificate or recovery key is lost, access to the encrypted data may also be lost.

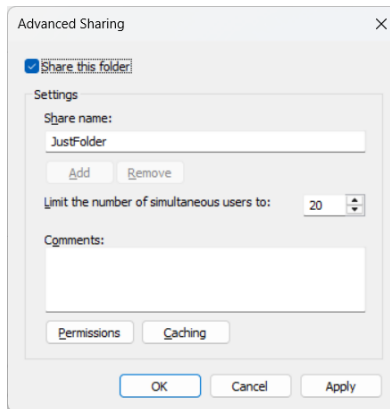
Folder Properties: TAB SHARING

The Sharing tab is used to make a folder available to other users over a network. It provides both a simplified sharing method and an Advanced Sharing option for more detailed configuration.

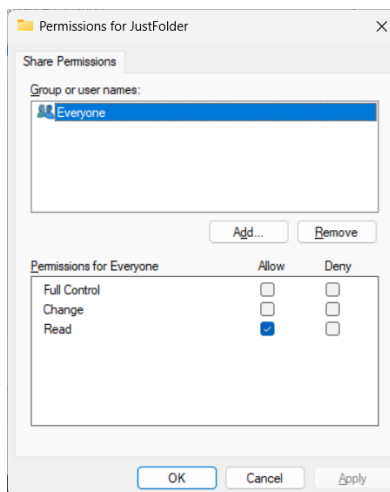


Share

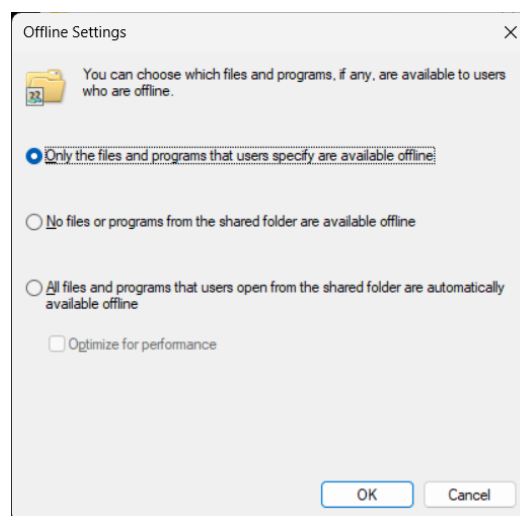




Advanced Sharing



Advanced Sharing > Permission Rights (Add or Remove users)



Advanced Sharing > Caching



Sharing a folder does not automatically give every user full access. Effective access depends on both:

- share permissions
- file-system permissions

When both permission sets apply, the *more restrictive effective permission* usually controls the user's access.

Quick Check:

A user has Full Control in the share permissions but only Read permission in the file-system permissions. Can the user modify the files?

OFFLINE SETTINGS EXPLAINED

This attribute allows users to access files even if they are not connected to the network.

Only the files and programs that users specify are available offline: Users must manually select which files to cache.

No files or programs from the shared folder are available offline: Disables offline file caching for the folder.

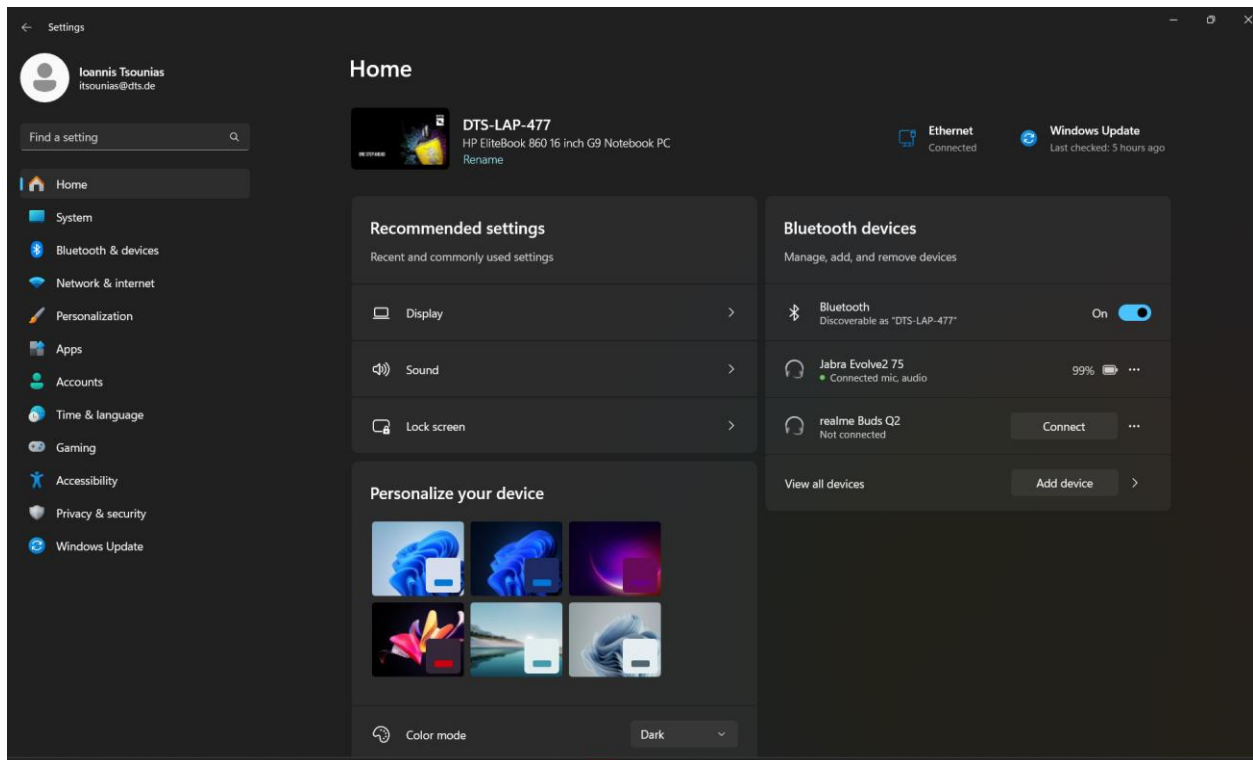
All files and programs that users open from the shared folder are automatically available offline: All files are cached automatically.



WINDOWS SETTINGS MENU – CONTROL PANEL

SETTINGS

The **Settings app** is one of the main places used to configure and manage Windows. It allows users and administrators to adjust areas such as display, sound, connected devices, network settings, personalization, accounts, privacy, security, and Windows Update.



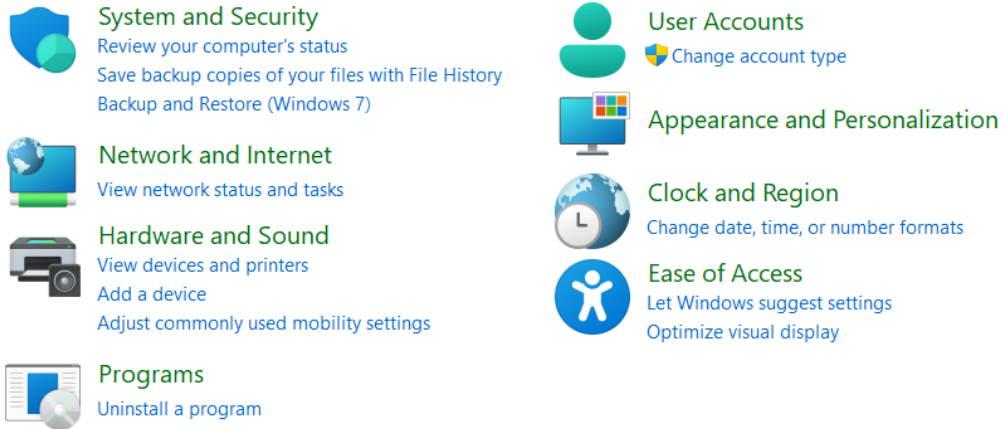
Settings Home (Win + I)



Control Panel is an older Windows management interface that is still used for several system and administrative tasks.

Adjust your computer's settings

View by: Category ▾



Control Panel menu

Although many options have moved to the Settings app, Control Panel is still commonly used for areas such as:

- network adapters
- user accounts
- installed programs
- power options
- system properties
- devices and printers
- administrative tools

Note:

In modern Windows versions, some options open in Settings, while others remain available only through Control Panel. For this reason, IT technicians should be familiar with both interfaces.



GROUP POLICIES – THE RULES BEHIND THE WINDOWS

Group Policy is a Windows management feature that allows administrators to define how computers and user accounts should operate. *Policies* can control security settings, system behavior, and access to features, desktop configuration, applications, and many other parts of the Windows environment.

On a standalone computer, *Local Group Policy* applies settings only to that device and its local users. In an Active Directory domain, Group Policy can be managed centrally and applied to selected computers and users across the organization.

Windows processes these policy settings during startup, sign-in, and regular background refresh intervals to help keep systems aligned with the intended configuration.

Inside the *Group Policy Editor*, there are two main branches: **Computer Configuration** and **User Configuration**. Although they may contain similar categories, they apply settings to different targets.

Computer Configuration

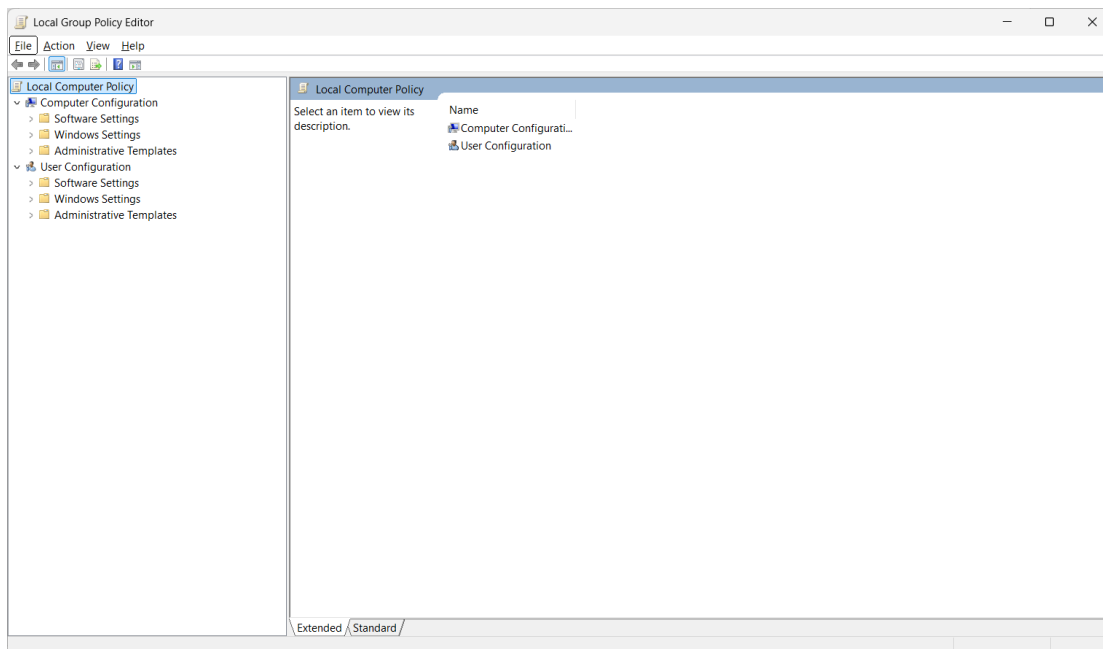
Computer Configuration contains *policies that apply to the device*, regardless of which user signs in. These settings can control areas such as security, Windows Update, firewall behavior, system services, startup scripts, and device restrictions.

For example, if a firewall policy is configured under Computer Configuration, it applies to the computer and affects users who work on that device.

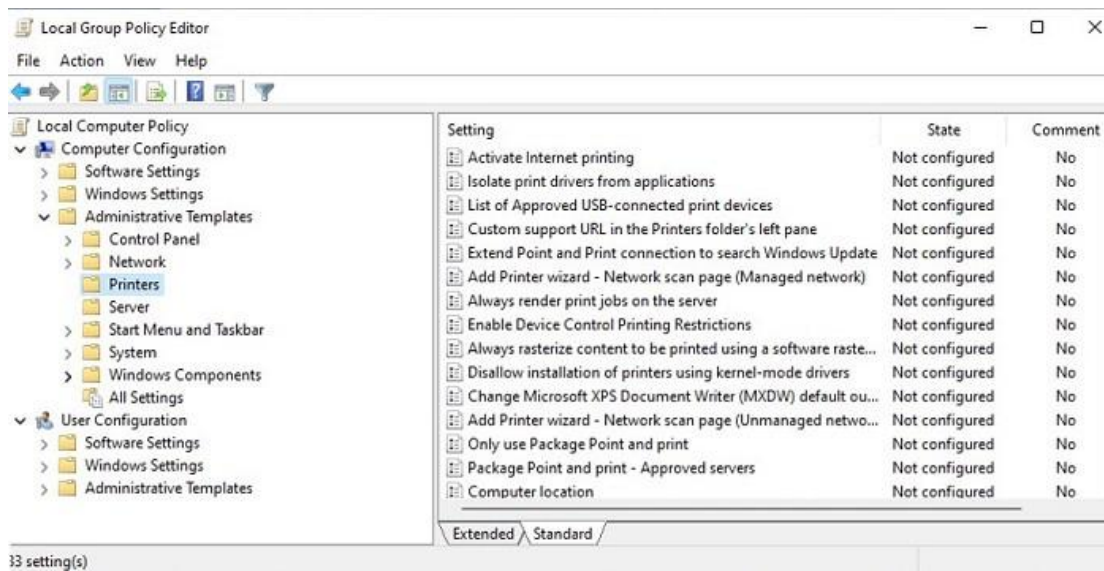
User Configuration

User Configuration contains *policies that apply to the user account*. These settings can control areas such as desktop restrictions, Start menu options, application settings, scripts, and Folder Redirection.

In an Active Directory environment, user policies can provide a consistent experience when the same user signs into different domain computers, depending on how the policies have been configured and applied.



Open Local Group Policy Editor: Press Win + R, enter gpedit.msc, and press Enter



Example of policy settings inside the Local Group Policy Editor

SECURITY & MAINTENANCE

CREATE A BACKUP

Backups are an essential part of system maintenance and data protection. They allow users and organizations to recover important information after accidental deletion, hardware failure, data corruption, malware, or another unexpected incident.

Windows provides different methods for protecting data. A **file backup** stores copies of selected files and folders, while a **system image** captures the Operating System, installed applications, system settings, and other required data so that the computer can be restored after a serious failure.

Cloud synchronization services such as OneDrive can also keep selected files available across multiple devices. However, synchronization and backup are not exactly the same. A complete backup strategy should include independent copies of important data and a clear recovery process.

Important principle:

A backup is useful only if the data can be restored successfully. Backups should therefore be checked and recovery procedures should be tested.

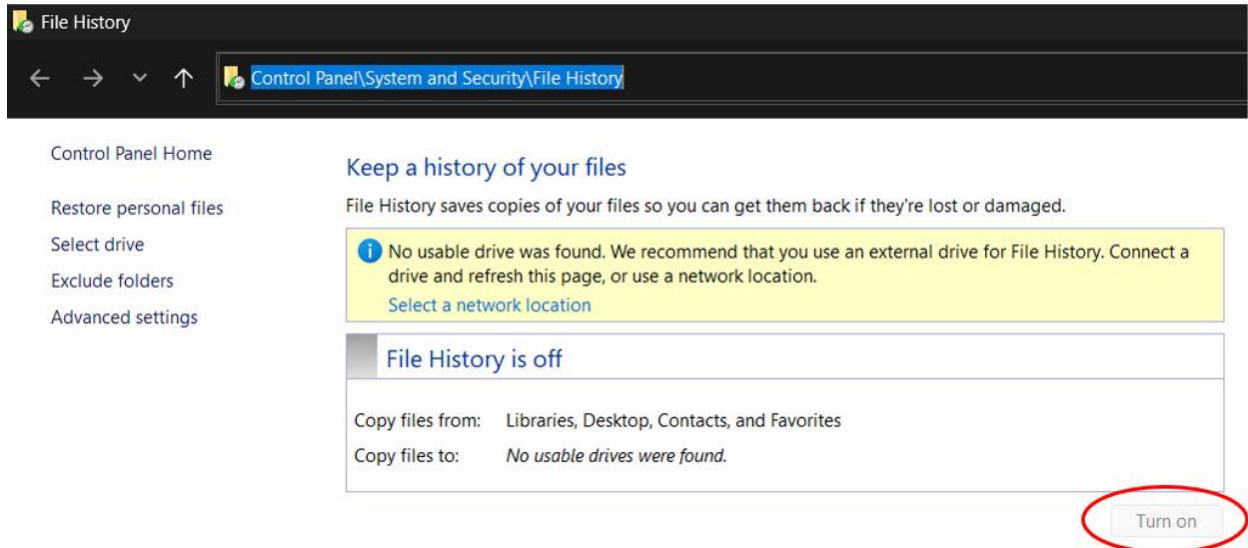
There are several ways to back up data in Windows. The following sections introduce some commonly used methods.

1. [File History](#)

File History creates regular copies of selected personal files and stores them on another drive or an available network location.

Requirement:

Connect or select a suitable backup destination before enabling File History.



[Control Panel > System and Security > File History \(Turn On\)](#)

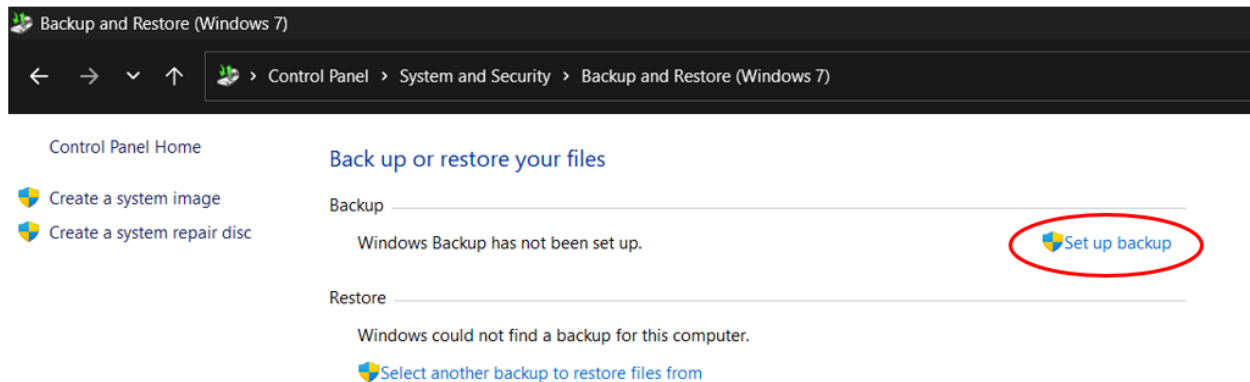


WINDOWS BACKUP & RESTORE

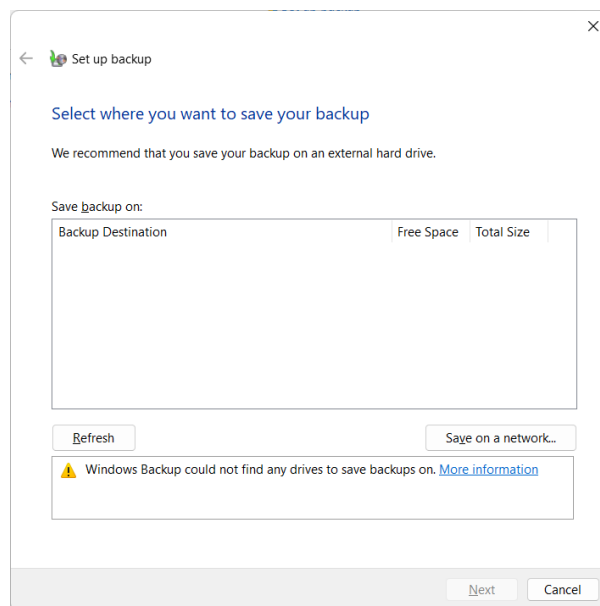
A backup can be stored on an external drive or an appropriate network location.

Important:

The backup destination should be separate from the physical drive that contains the original data whenever possible.



Control Panel > System and Security > Backup and Restore (Windows 7) > Set up Backup



Select the backup destination, verify that sufficient storage space is available, and then select **next**



SYSTEM IMAGE VS FILE BACKUP

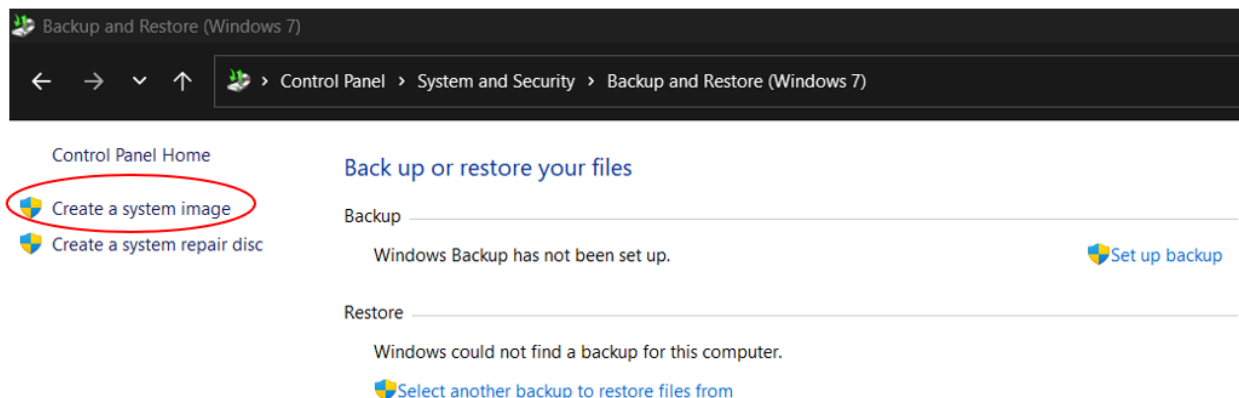
A **system image** is a complete backup of the system volumes required for Windows to operate. It can include the Operating System, system settings, installed applications, drivers, and user data.

A system image is mainly used to restore a computer after a serious failure, such as disk failure, severe system corruption, or an unsuccessful system change. During a system-image recovery, the selected system volumes are restored as a complete unit rather than as individual files.

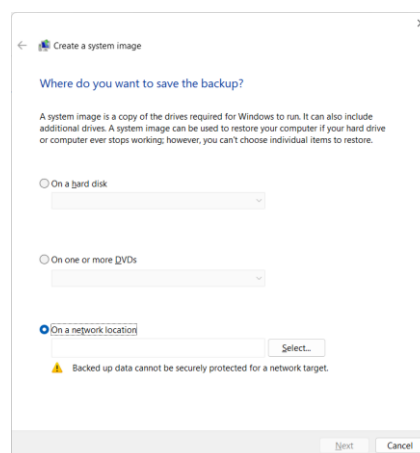
A **file backup**, on the other hand, stores copies of selected files and folders. It is normally faster, requires less storage space, and allows individual items to be restored without recovering the entire system.

The correct backup method depends on the recovery requirement. File backups are suitable when individual documents or folders must be recovered, while a system image is more suitable when the complete Windows environment must be restored.

Creating a system image



Control Panel > System and Security > Backup and Restore (Windows 7) > Create a system image



Select the backup location > Next > Review the selected drives > Start backup



BASIC WINDOWS MAINTENANCE – FREEING UP STORAGE SPACE

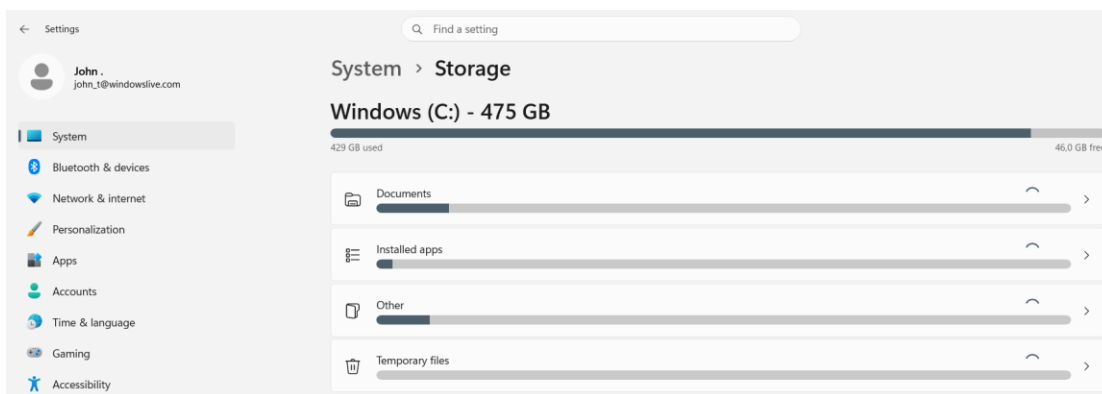
Low available disk space can affect system performance, application operation, and the installation of Windows updates.

Windows provides several tools for identifying and removing unnecessary temporary files. Before deleting any data, users should review the selected categories and make sure that important personal or business files are not included.

Clear Temp Files

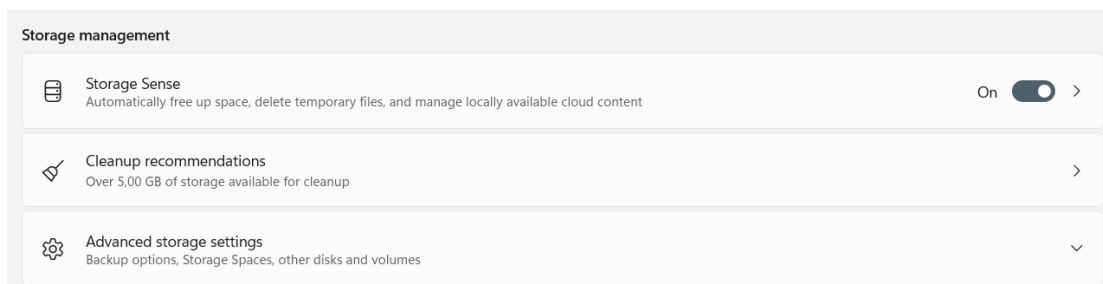
METHOD 1 – TEMPORARY FILES AND CLEANUP RECOMMENDATIONS

Open:



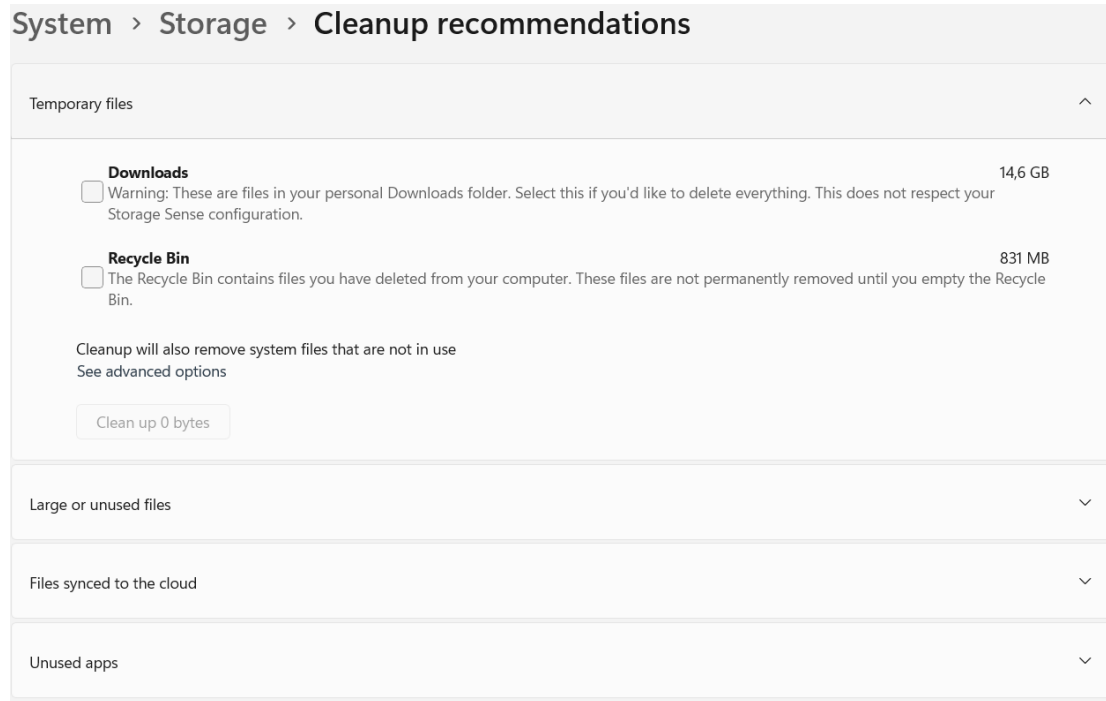
[Start > Settings > System > Storage](#)

Select **Temporary files** or **Cleanup recommendations**.



Windows displays categories of files that may be removed and the amount of storage space that can be recovered.





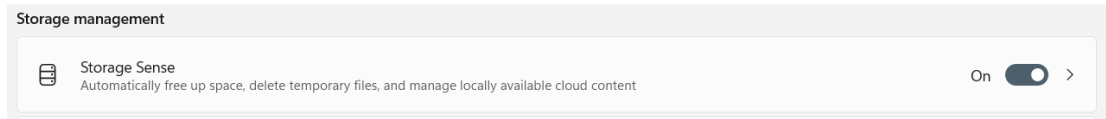
Review the selected categories carefully before choosing **Remove files** or **Clean up**. Items such as *Downloads*, *Recycle Bin* contents, and previous *Windows installations* should not be removed unless the user has confirmed that they are no longer required!

METHOD 2 – STORAGE SENSE

Open:

[Start > Settings > System > Storage > Storage Sense](#)

Storage Sense can automatically remove unnecessary temporary files and manage files stored in the Recycle Bin.



Its settings should be reviewed before activation. Files in the Downloads folder and locally available cloud content should not be managed automatically unless this matches the user's or organization's requirements.

Storage Sense normally manages the Windows system drive, which is usually the C: drive.

System > Storage > Storage Sense

Cleanup of temporary files

☒ Keep Windows running smoothly by automatically cleaning up temporary system and app files

Automatic User content cleanup

☒ On

Storage Sense runs when disk space is low. We clean up enough space to help your system run its best. We cleaned up 0 bytes of space in the past month. Last run: 19/10/2024 13:54.

Configure cleanup schedules

Run Storage Sense

During low free disk space (default) ▾

Delete files in my recycle bin if they have been there for over:

30 days (default) ▾

Delete files in my Downloads folder if they haven't been opened for more than:

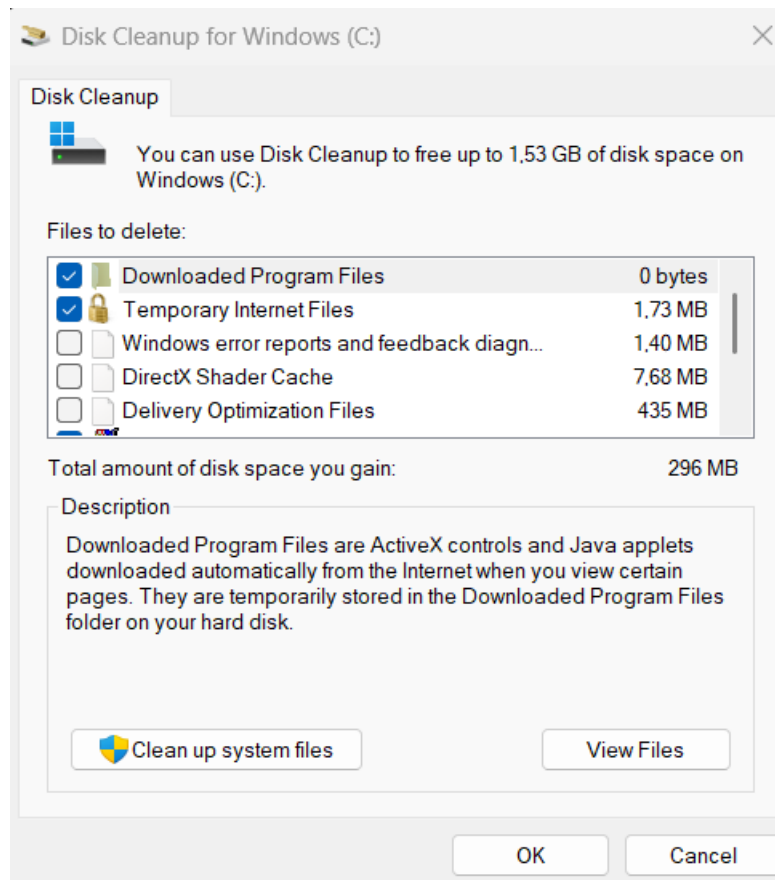
Never (default) ▾

Run Storage Sense now



METHOD 3 – DISK CLEANUP

Search for **Disk Cleanup** from the **Start menu**.



Select the drive that must be checked and review the available file categories.

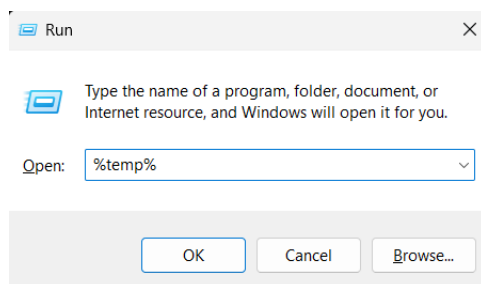
To identify additional system files that may be removed, select *Clean up system files*.

Do not select a category only because Windows displays it. Read its description and confirm that the files are no longer required before deleting them.

METHOD 4: OPTIONAL MANUAL CHECK

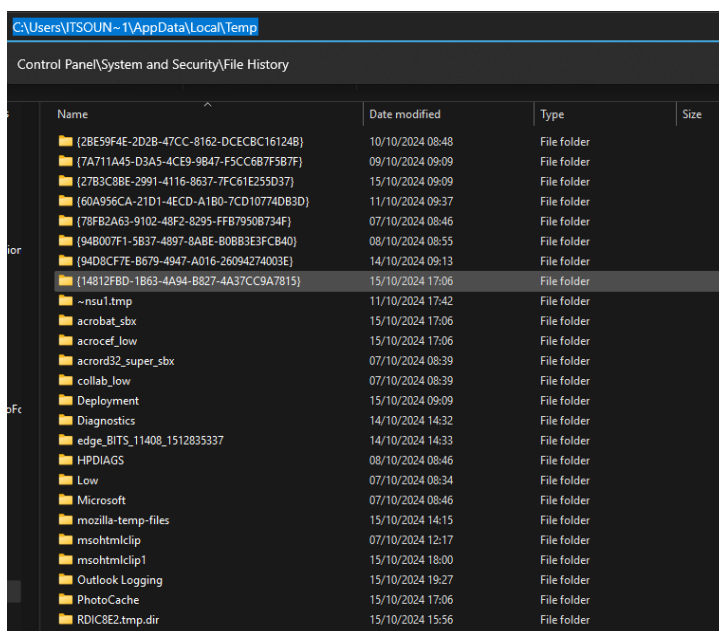
Manually Deleting Temp Files

Open Run Dialog Box (Win+R)



%temp% > OK

Temporary files that are no longer required may be removed. Some files may currently be used by Windows or by an application and cannot be deleted. These files should be skipped.



Choose what to Delete

Manual deletion should not replace the Windows storage-management tools or a proper investigation of the cause of low disk space.



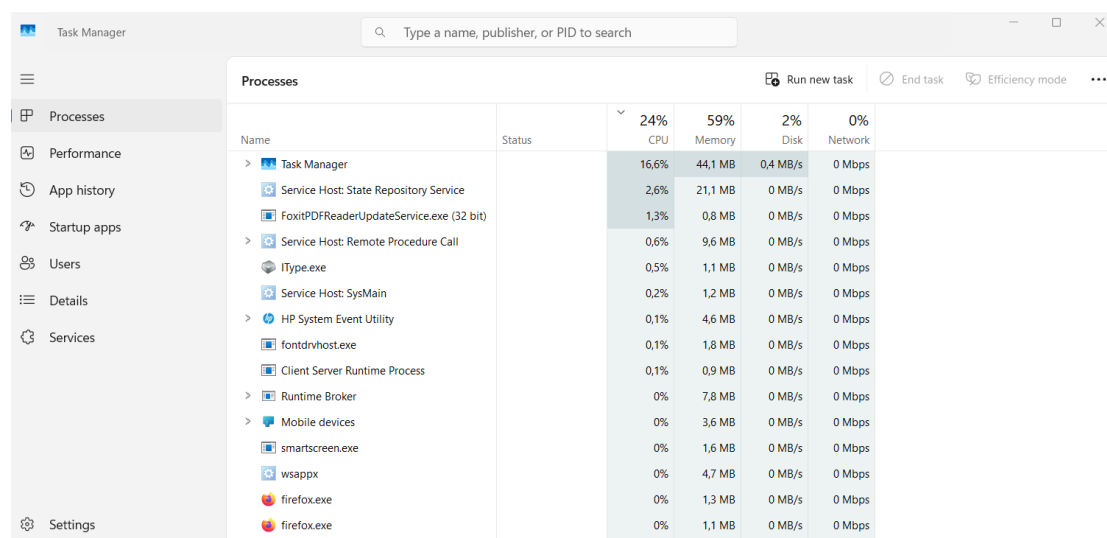
MONITORING & TROUBLESHOOTING TOOLS IN WINDOWS OS

Windows includes several built-in tools that allow technicians to monitor system performance, investigate problems, and collect troubleshooting evidence.

Task Manager

Task Manager is a built-in Windows tool used to view running applications and processes, monitor system-resource usage, manage startup applications, and perform basic troubleshooting actions.

It provides live information about CPU, memory, disk, network, and graphics-resource usage.



Task Manager Default Page

Task Manager section	Purpose
Processes	Displays running applications and background processes together with their current CPU, memory, disk, network, and GPU usage.
Performance	Provides live graphs and an overview of CPU, memory, disk, network, and GPU performance.
App History	Displays historical resource usage for supported applications.
Startup apps	Displays applications configured to start when a user signs in and their estimated startup impact.
Users	Displays signed-in users and the resources being used by their processes.
Details	Provides detailed information and additional management options for running processes.
Services	Displays Windows services and their current status. The Services console provides more complete service-management options.

Task Manager can be used to end an application that is no longer responding. However, before ending a task or process, the technician should identify it and consider the possible impact.



Event Viewer

Event Viewer is a Windows tool used to view events recorded by the Operating System, applications, services, and security components.

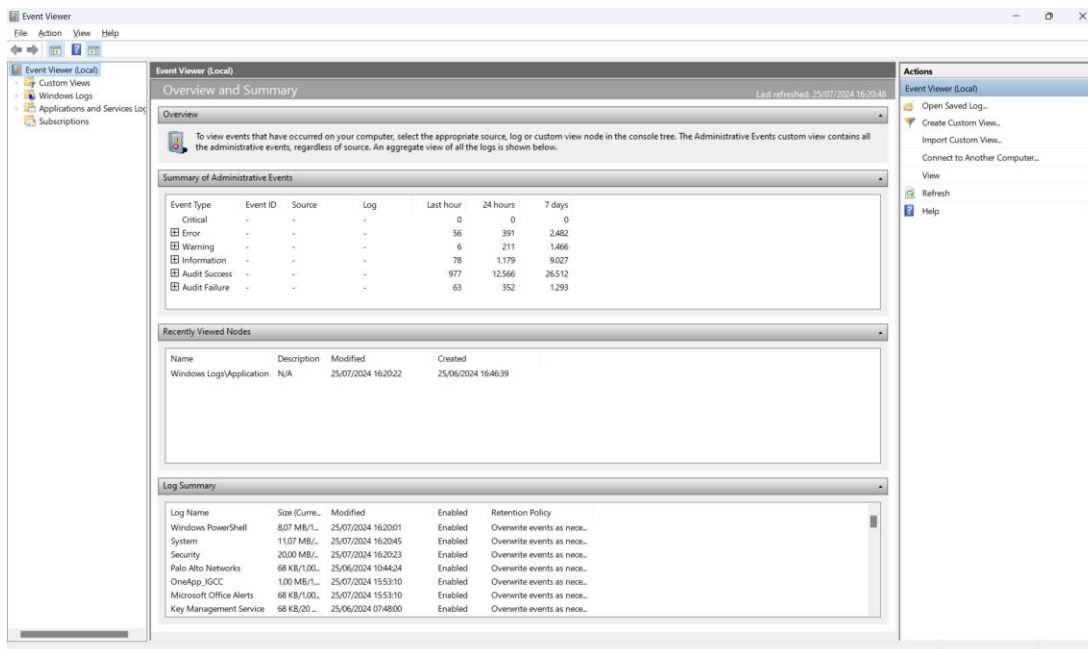
Common Windows logs include:

- Application
- Security
- Setup
- System
- Forwarded Events

Each event may contain information such as its date and time, source, Event ID, severity level, and description.

Event Viewer is useful when a technician needs to investigate application failures, system errors, service problems, security events, installation issues, or other recorded system activity.

Warnings and errors should be treated as troubleshooting evidence, but they do not automatically identify the root cause of a problem. The technician should correlate the event with the reported symptoms, the time at which the issue occurred, and evidence collected from other tools.



Event Viewer Default Page



Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - System
 - Forwarded Events
 - Applications and Services Logs
 - Subscriptions

Application Number of events: 6307

Level	Date and Time	Source	Event ID	Task Category
Information	25/07/2024 16:20:19	Outlook	63	None
Information	25/07/2024 16:19:52	Security-SPP	16384	None
Information	25/07/2024 16:18:52	Security-SPP	16394	None
Information	25/07/2024 16:07:29	Security-SPP	16384	None
Information	25/07/2024 16:06:59	Security-SPP	16394	None
Information	25/07/2024 15:44:17	HotKeyServiceUWP	0 (2)	
Information	25/07/2024 15:36:34	HotKeyServiceUWP	0 (2)	
Information	25/07/2024 15:20:56	Security-SPP	16384	None
Information	25/07/2024 15:20:25	Security-SPP	16394	None
Information	25/07/2024 15:20:05	Security-SPP	16384	None
Information	25/07/2024 15:19:33	Security-SPP	16394	None
Information	25/07/2024 15:07:24	Security-SPP	16384	None
Information	25/07/2024 15:06:53	Security-SPP	16394	None
Warning	25/07/2024 14:49:35	SoeCl	1202	None
Information	25/07/2024 14:42:10	Security-SPP	16384	None
Information	25/07/2024 14:41:38	Security-SPP	16394	None
Information	25/07/2024 14:20:29	Security-SPP	16384	None
Information	25/07/2024 14:19:59	Security-SPP	16394	None
Information	25/07/2024 14:19:54	Security-SPP	16384	None
Information	25/07/2024 14:19:22	Security-SPP	16394	None
Information	25/07/2024 14:15:05	HotKeyServiceUWP	0 (2)	

Event 63, Outlook

General Details

The Exchange web service request GetAppManifests succeeded.

Log Name: Application
 Source: Outlook
 Event ID: 63
 Level: Information
 User: N/A
 OpCode: Info
 More Information: [Event Log Online Help](#)

Logged: 25/07/2024 16:20:19
 Task Category: None
 Keywords: Classic
 Computer: DTS-LAP-477.dts-systeme.intra

Actions

- Application
- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- Attach a Task To this Log...
- View
- Refresh
- Help
- Event 63, Outlook
 - Event Properties
 - Attach Task To This Event...
 - Copy
 - Save Selected Events...
 - Refresh

Windows Logs



RESOURCE MONITOR

Resource Monitor provides *more detailed real-time information* than Task Manager about how individual processes use system resources.

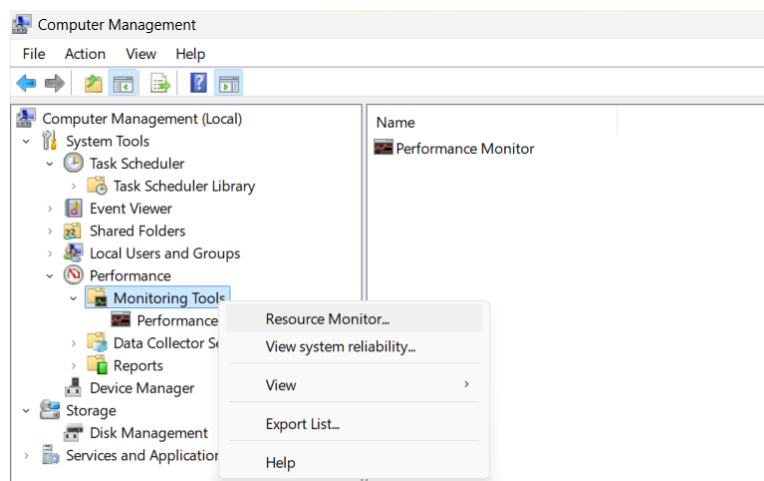
It provides detailed information about:

- CPU activity
- Physical memory usage
- Disk activity
- Network activity

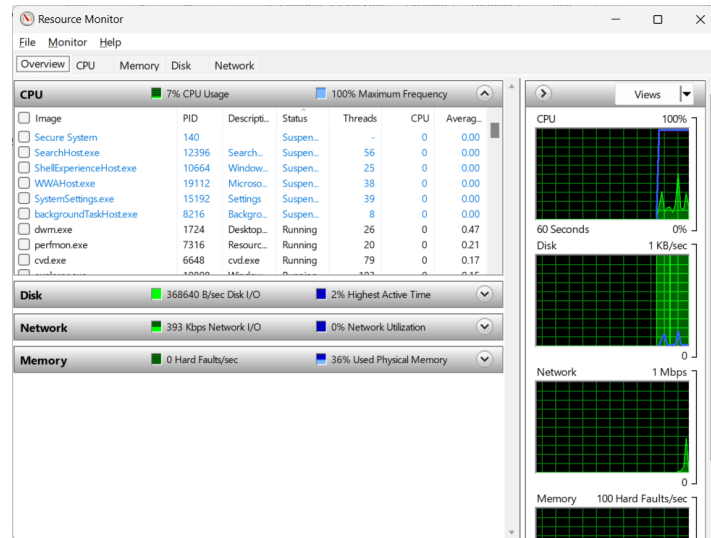
Resource Monitor is useful when a technician needs to identify which process is generating high CPU usage, consuming memory, continuously accessing the disk, or producing network traffic.

Open Resource Monitor:

1. Search for Resource Monitor from the Start menu
2. Press Win + R, enter “resmon”, and press Enter
3. Open Task Manager > Performance and select Resource Monitor, where available



Start > Computer Management > System Tools > Performance R. click: Resource Monitor



Resource Monitor Overview

RELIABILITY MONITOR

Reliability Monitor provides a *chronological view* of system stability, failures, and important system changes.

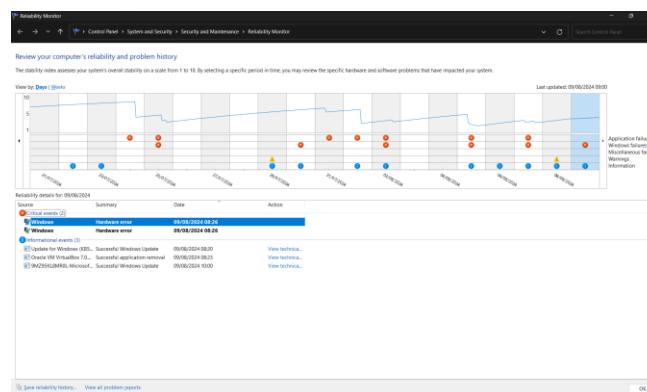
It can display information such as:

- Application failures
- Windows failures
- Hardware-related events
- Failed installations
- Windows updates
- Application installations

Reliability Monitor is especially useful when a user reports that a problem began after a recent update, application installation, driver change, or system failure.

[Open Reliability Monitor:](#)

Search for View reliability history from the Start menu



Press Win + R, enter perfmon /rel, and press Enter



Tool	Best Used for
Task Manager	Quick live overview of applications, processes, users, and resource usage
Resource Monitor	Detailed live analysis of CPU, memory, disk, and network activity by process
Event Viewer	Examination of recorded Operating System, application, service, and security events
Reliability Monitor	Chronological review of failures, installations, updates, and system changes

Windows Memory Diagnostic Tool

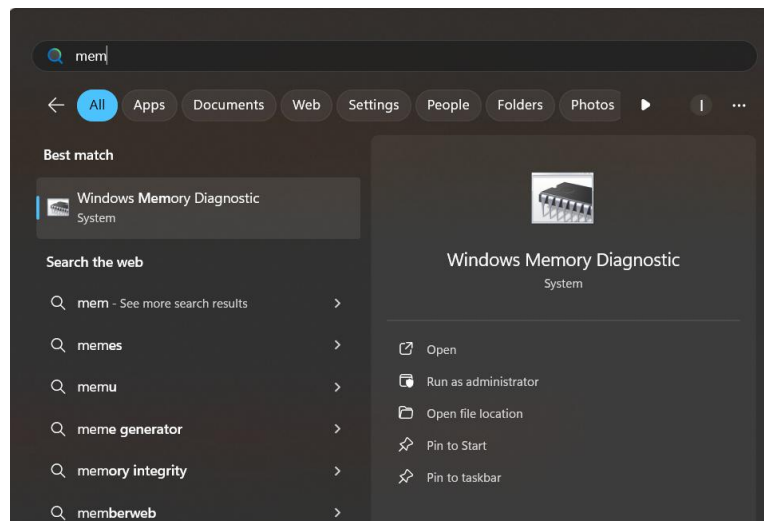
Windows Memory Diagnostic is a built-in troubleshooting tool that tests the computer's physical memory for possible errors.

The test requires a restart because the memory must be checked outside the normal Windows working environment.

To start the tool:

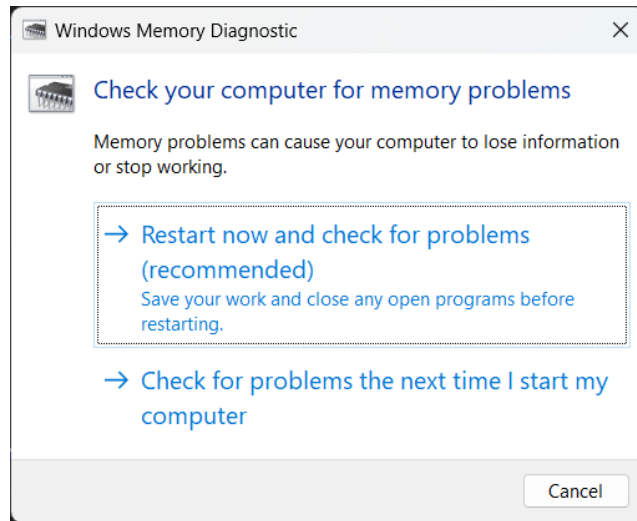
Search from the Start: Windows Memory Diagnostic or press Win + R and enter:

"mdsched.exe"



Memory Diagnostic Tool





Windows Memory Diagnostic

QUICK CHECK – SELECT THE APPROPRIATE WINDOWS TOOL

For each situation, identify the Windows tool that you would use first. Explain your choice briefly:

1. An application is no longer responding, and CPU usage appears unusually high.
2. A user reports that system crashes started three days ago after a software installation.
3. You need detailed information about which process is continuously writing data to the disk.
4. You need to locate a specific system error that was recorded at approximately 10:15 am.
5. You suspect that the workstation may have faulty physical memory.

Windows Command Line Usage

Windows provides *command-line* tools that allow users and administrators to perform tasks, retrieve system information, and collect troubleshooting evidence without navigating through multiple graphical menus.

Command Prompt can be used for file and folder management, system inspection, process monitoring, network troubleshooting, and administrative operations.

Some commands can be executed by a standard user, while others require an elevated Command Prompt with administrative privileges. Commands that modify processes, disks, system files, or system configuration should be used only after their purpose and possible impact have been understood.

Basic CMD Commands

Command	Purpose
cd	Displays the current directory or changes the current working directory
cd \	Navigates to the root directory of the current drive
cd..	Moves one level up in the folder structure
dir	Displays the files and subdirectories contained in the current or specified directory
Mkdir or md	Creates a new directory
tree	Displays the folder structure of a drive or directory in a tree-like format.
Copy	Copies one or more files to another location.
rename (or ren)	Renames a file or directory.
systeminfo	Displays detailed information about the computer, Operating System, hardware, memory, network adapters, and system configuration
winver	Opens the <i>About Windows</i> dialog box and displays the installed Windows version and build information
tasklist	Displays a list of processes currently running on the local computer or a specified remote computer
taskkill	Ends a process by using its image name, Process ID, or another filter



Examples:

1. *Display the contents of the current directory:*

Command: dir

2. *Create a new folder:*

Command: Mkdir *Project_Files*

3. *Display detailed system information:*

Command: systeminfo

4. *Display running processes:*

Command: tasklist

SYSTEM INFORMATION AND TROUBLESHOOTING COMMANDS

Command	Purpose
hostname	Displays the host-name portion of the computer's full name
getmac	Displays the MAC addresses and associated transport information for the computer's network adapters
perfmon	Opens Windows Performance Monitor. Additional parameters can open Resource Monitor, Reliability Monitor, or a system diagnostic report
resmon	Opens Resource Monitor
eventvwr	Opens Event Viewer

Examples for **perfmon**:

1. Open Resource Monitor:

Command: perfmon /res

2. Open Reliability Monitor:

Command: perfmon /rel

3. Start a system diagnostic report:

Command: perfmon /report



ADMINISTRATIVE AND SYSTEM REPAIR COMMANDS

Command	Purpose	Important Note
sfc /scannow	Scans all protected Windows system files, verifies their integrity, and replaces incorrect or damaged protected files when a valid replacement is available.	Requires an elevated Command Prompt. It checks protected Windows system files, not personal files or every application file.
chkdsk	Checks the file system and file-system metadata of a selected volume for logical and physical errors.	Without repair parameters, it reports the status of the volume but does not repair errors. Parameters such as <i>/f</i> and <i>/r</i> may require administrative privileges, exclusive access, or a restart.
diskpart	Opens the DiskPart command interpreter, which is used to view and manage disks, partitions, volumes, and virtual disks.	Some DiskPart commands can delete partitions, remove volumes, or make data inaccessible. Use it only after positively identifying the correct disk.
list disk	Displays the disks detected by Windows together with information such as disk number, status, size, available space, and partition style.	This is a DiskPart command. Enter <i>diskpart</i> first, run <i>list disk</i> , and then use <i>exit</i> to leave DiskPart.

IMPORTANT SAFETY PRINCIPLE

Administrative and **system-repair** commands should not be executed randomly. Before running a command, the technician should understand its purpose, identify the system object that will be affected, and consider the possible impact on data and system availability.

Before using a command that can modify system files, processes, disks, partitions, or volumes, the technician should:

- confirm whether administrative privileges are required
- identify the correct disk, volume, process, or system component
- collect and record the relevant troubleshooting evidence
- consider whether a backup or recovery method is required
- understand the parameters being used
- define how the result will be verified

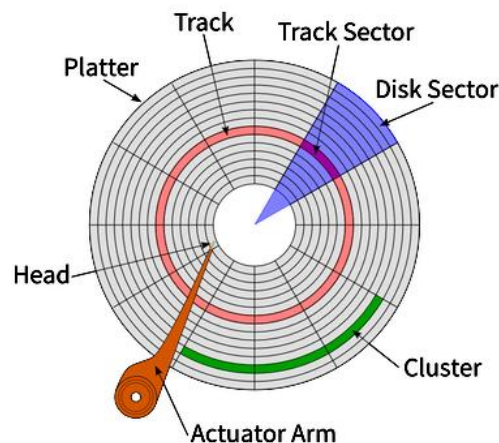
When an incident is escalated, the technician should **document** the commands executed, the output received, any changes made, and the current condition of the system.

DISK FORMATTING AND VOLUME PREPARATION

Before Windows can normally store files on a new volume, the volume must be prepared with a file system. This process is known as formatting. Formatting creates the file-system structures that Windows uses to organize, locate, store, and manage files and folders on the selected volume. Common Windows file systems include NTFS, exFAT, FAT32, and, in specific server and storage scenarios, ReFS.

A storage device is divided into *physical* or *logical* sectors. The file system groups sectors into allocation units, which are commonly known as *clusters*. A cluster is the smallest amount of storage space that the file system normally allocates to a file.

Formatting is performed on a selected volume or partition. It is not the process that physically creates the tracks and sectors of a modern storage device.



HDD



Formatting should not be used as a general troubleshooting response to every Windows problem.

It is normally performed when preparing a new or reused volume, changing the file system, or carrying out a planned clean Operating System installation.

Formatting removes access to the existing files on the selected volume and may overwrite existing data, depending on the selected formatting method. Important data must therefore be backed up and verified before the operation begins.

CHECKS BEFORE FORMATTING OR REINSTALLING WINDOWS

Before formatting a volume or performing a clean Windows installation, the technician should:

- confirm the identity of the correct disk, partition, and volume
- back up all required personal and business data
- verify that the backup can be accessed
- check whether encryption or recovery keys must be saved
- identify the installed Windows edition and activation method
- prepare suitable Windows installation media when required
- confirm that the required device drivers and network connectivity will be available
- document the existing configuration, applications, and user requirements
- prepare a recovery or rollback plan where possible

A technician should **never** select a disk or volume based only on its drive letter or disk number. Its capacity, contents, label, partition layout, and purpose should also be verified.

COMMON REASONS FOR FORMATTING A VOLUME

Formatting may be required in situations such as:

- preparing a new or previously unallocated volume for data storage
- changing the file system of a non-system volume
- reusing a storage device after its data has been backed up or is no longer required
- preparing a volume as part of a planned clean Operating System installation
- removing an existing volume structure in a controlled deployment or decommissioning process
- correcting a file-system configuration that cannot be resolved through an appropriate and less disruptive method

Poor performance alone is not sufficient evidence that formatting is required. The technician should first investigate storage capacity, startup applications, system-resource usage, hardware condition, updates, malware, and recorded system errors.

Similarly, formatting should not automatically be treated as the first response to malware. The correct action depends on the type of infection, the organization's security procedures, the sensitivity of the system, and the available recovery options.

Formatting method	Description	Use and limitations
Quick Format	Creates new file-system structures, including the file table and root directory, without carrying out a sector-by-sector scan of the volume	Faster and suitable when the volume is known to be operational. Existing data is not securely erased and may remain recoverable until overwritten.
Full Format	Creates the selected file system and performs a more complete examination of the volume. It takes considerably longer than a quick format	May be appropriate when the condition of the volume must be examined more thoroughly. It is not a repair for failing hardware and does not replace drive-health diagnostics or backups

Important:

Formatting creates the selected file system, such as NTFS or exFAT. It does not always create a FAT file table.

A quick format should not be considered a secure data-erasure method. A full format should also not be presented as a guaranteed method of repairing damaged storage hardware.



FORMATTING AN EXISTING **NON-SYSTEM** VOLUME

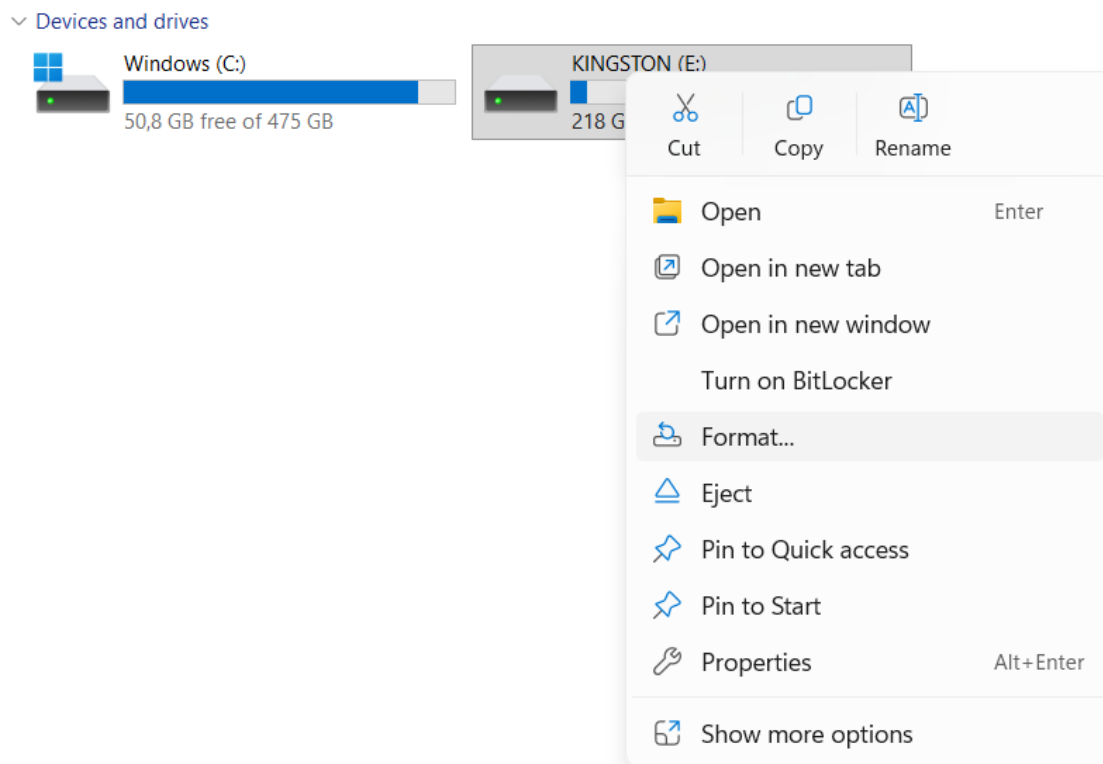
Before proceeding, confirm that the selected volume **does not contain** required data and is not the active Windows system volume.

[Open](#):

File Explorer > This PC

Right-click the appropriate non-system volume or removable drive and select:

Format



[This PC > Right Click on Disk: > Format](#)

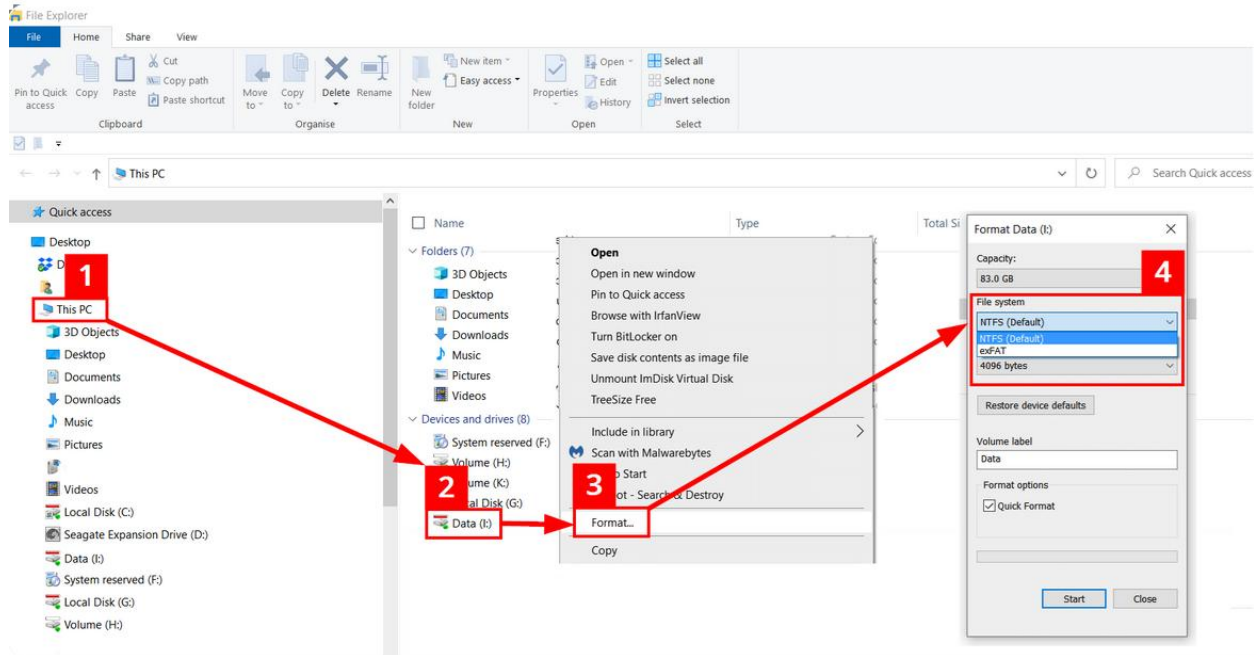
Review the following settings:

- volume capacity
- file system
- allocation unit size
- volume label
- Quick Format selection

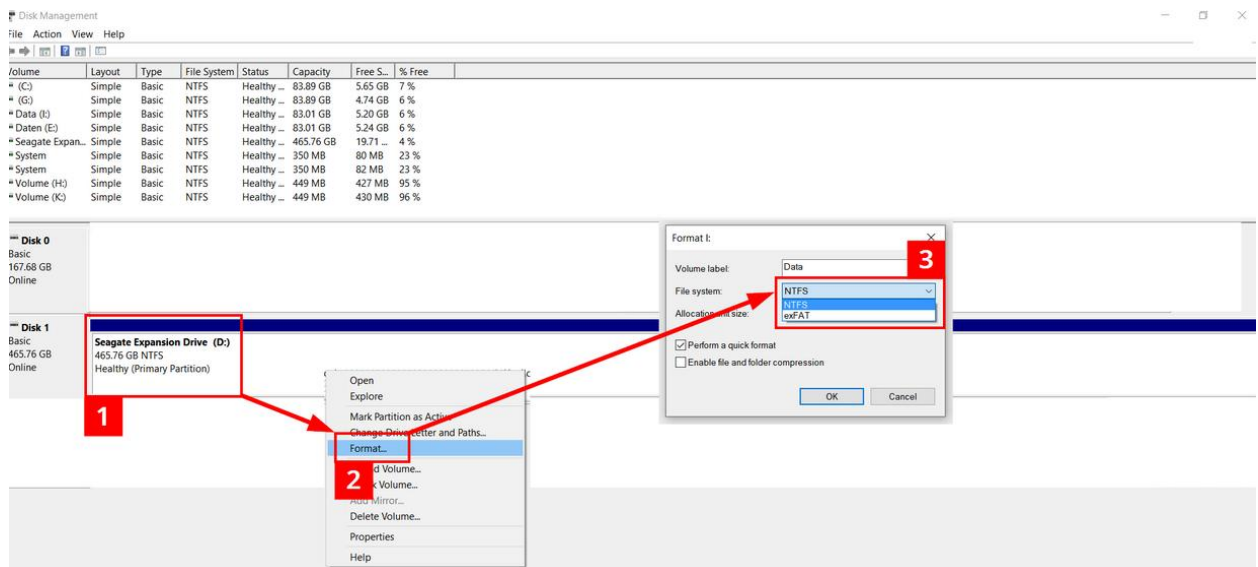
Confirm again that the correct volume has been selected before starting the operation.

The Windows system volume cannot be formatted while it is being used by the running Operating System.





OR



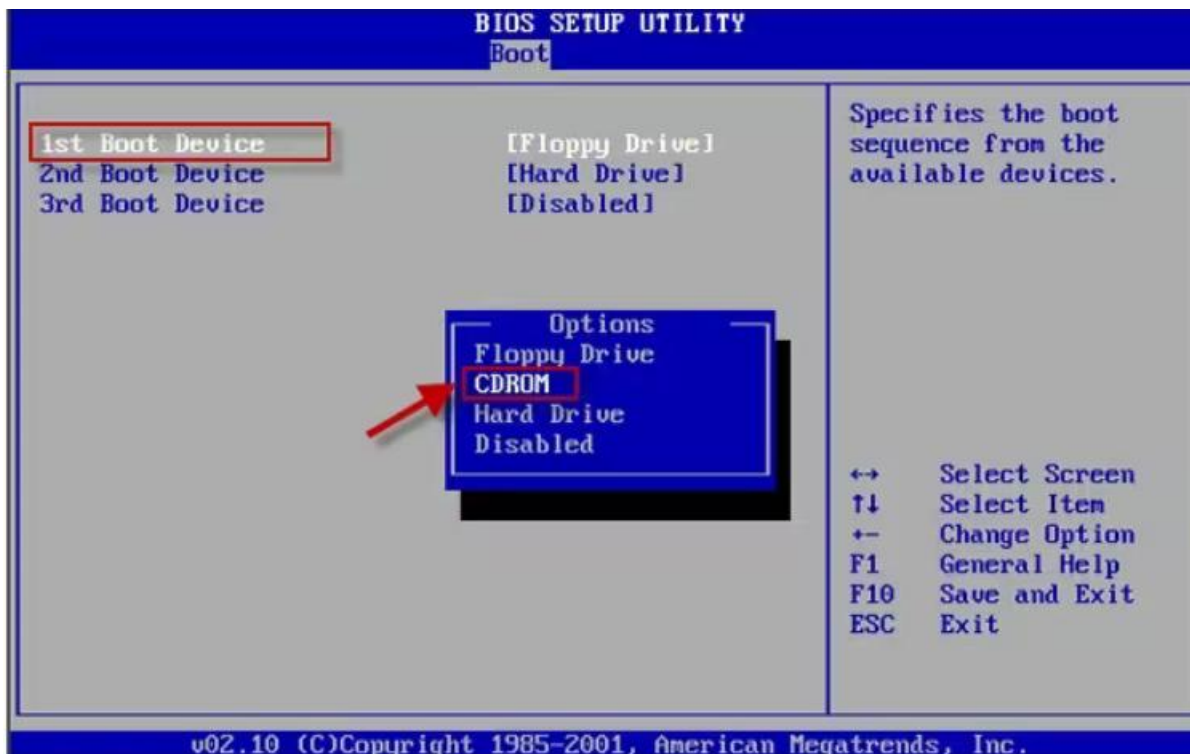
INSTALLING OR REINSTALLING WINDOWS FROM INSTALLATION MEDIA

To perform a clean Windows installation, the computer may need to start from Windows installation media, normally a bootable USB drive.

Depending on the device, the technician may use a temporary boot menu or change the boot order in the UEFI or legacy BIOS firmware settings. The correct key and menu vary between manufacturers.

The firmware does not format the disk. It only starts the computer from the selected installation or recovery media.

Change the Booting Sequence from BIOS

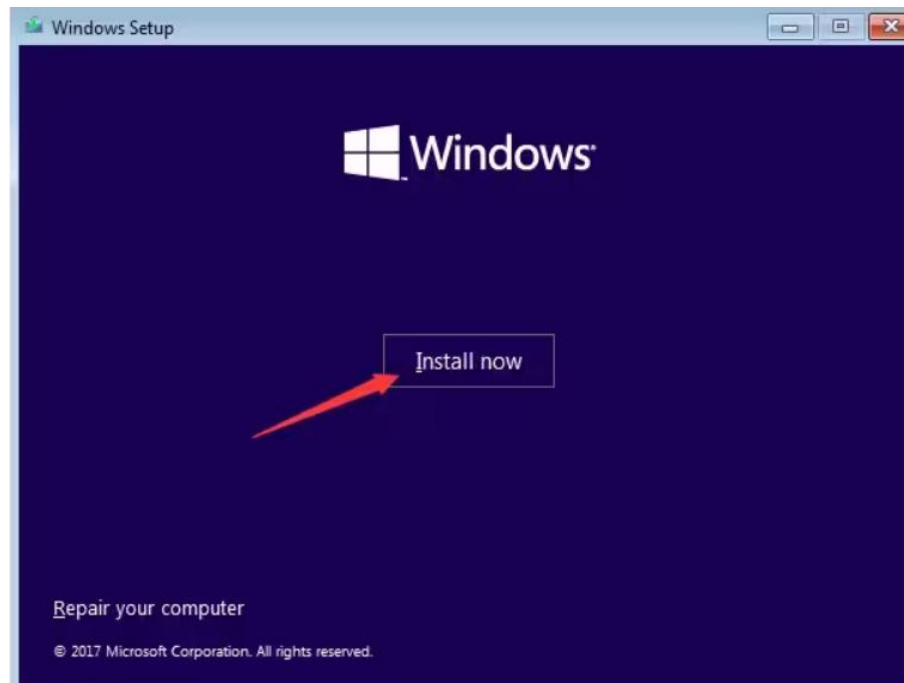


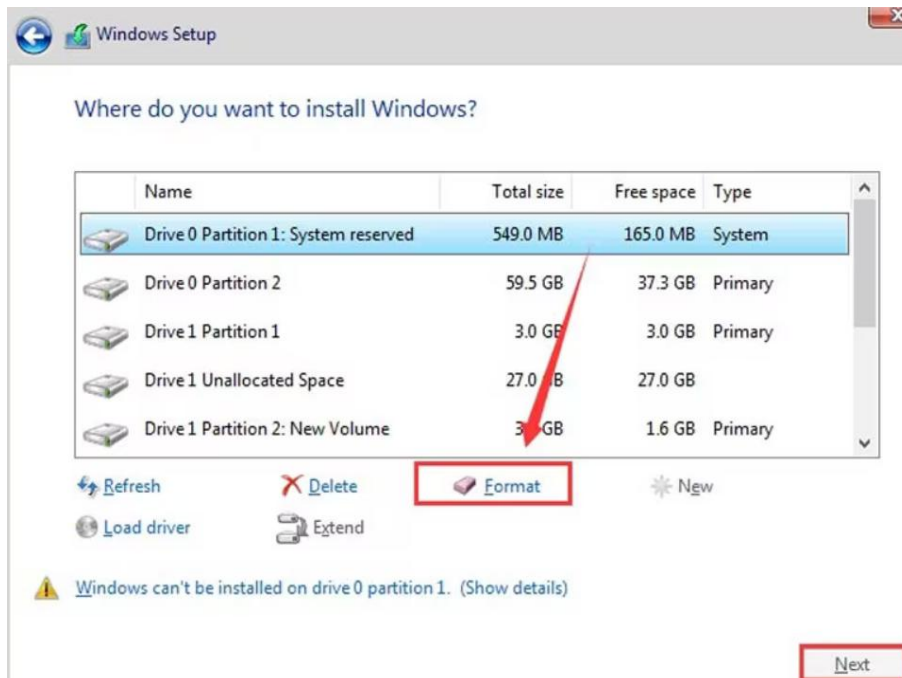
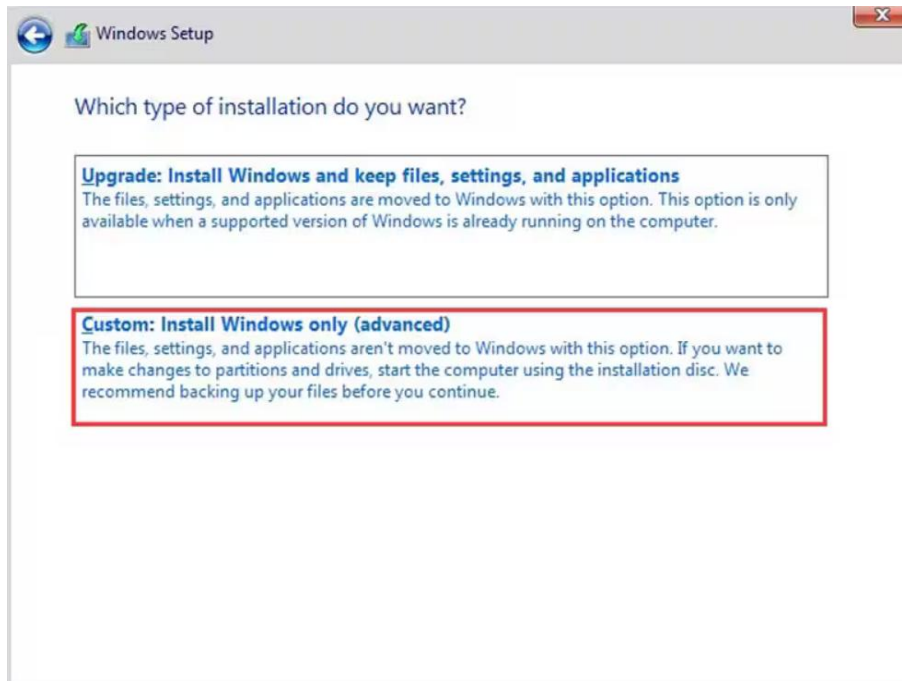
GENERAL CLEAN-INSTALLATION PROCESS

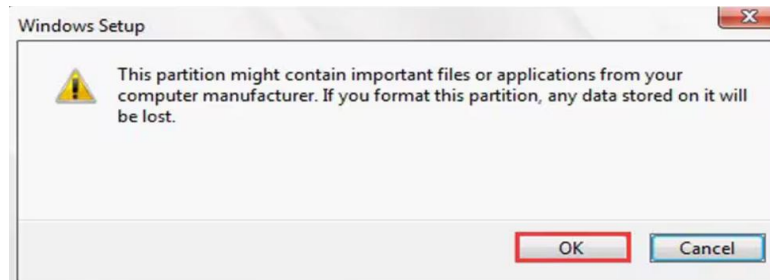
After starting the computer from the installation media:

- Select the required language, regional, and keyboard settings.
- Begin the Windows installation process.
- Select the Windows edition that matches the available license or digital entitlement.
- Review and accept the applicable license terms.
- Select the clean or custom installation option when a completely new installation is required.
- Carefully identify the correct destination disk and partition.
- Modify or delete partitions only when this is part of the approved installation plan and all required data has been backed up.
- Select the required destination and begin the installation.
- Complete the Windows initial configuration after the computer restarts.
- Verify activation, updates, drivers, security settings, network connectivity, applications, and access to required data.

A clean installation removes applications, settings, and data from the selected installation environment. It should therefore be performed only after the destination disk and recovery requirements have been confirmed.







We follow the steps it asks us until the whole process is completed!

PREPARING A NEW DISK IN WINDOWS

A newly added internal or virtual disk may not appear immediately in File Explorer. Before it can normally store files, it may need to be initialized and configured through Disk Management.

Open:

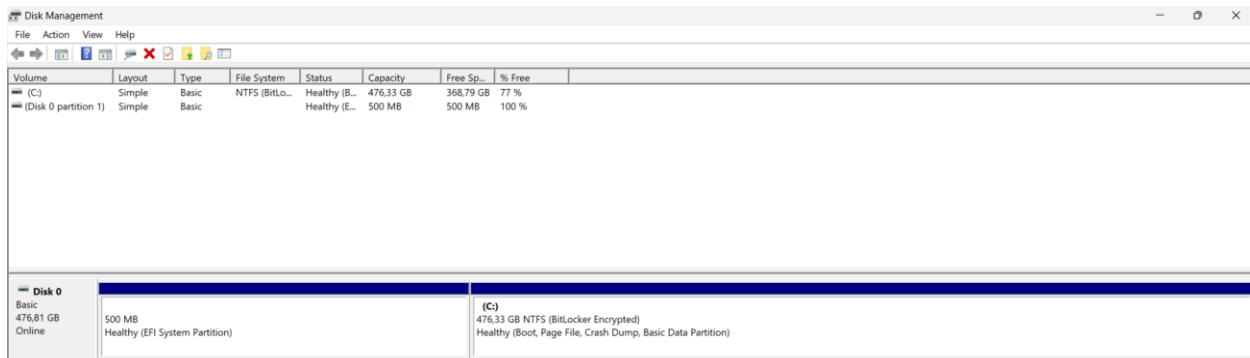
Start > Search for "Create and format hard disk partitions"

or:

Computer Management > Storage > Disk Management

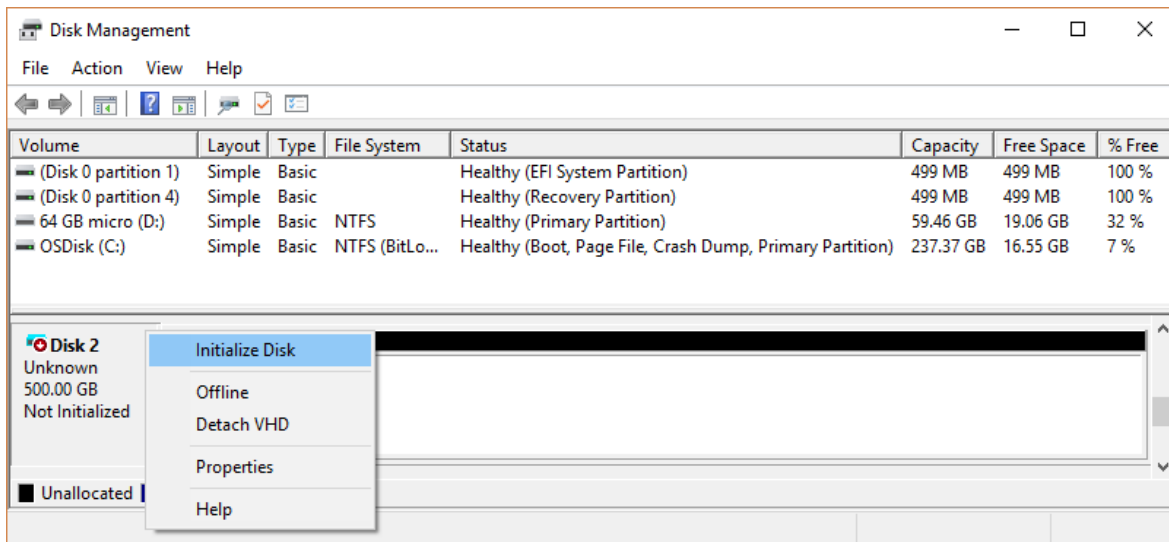
In Disk Management:

1. Identify the new disk by checking its disk number, capacity, and status.
2. If required, bring the disk Online.
3. Right-click the disk label area and select Initialize Disk.
4. Confirm that the correct disk is selected.
5. Select the appropriate partition style, normally GPT for a modern system.
6. Right-click the unallocated space and select New Simple Volume.
7. Specify the volume size.
8. Assign a drive letter or mount point.
9. Select the required file system and volume label.
10. Complete the process and verify that the volume appears correctly in Disk Management and File Explorer.

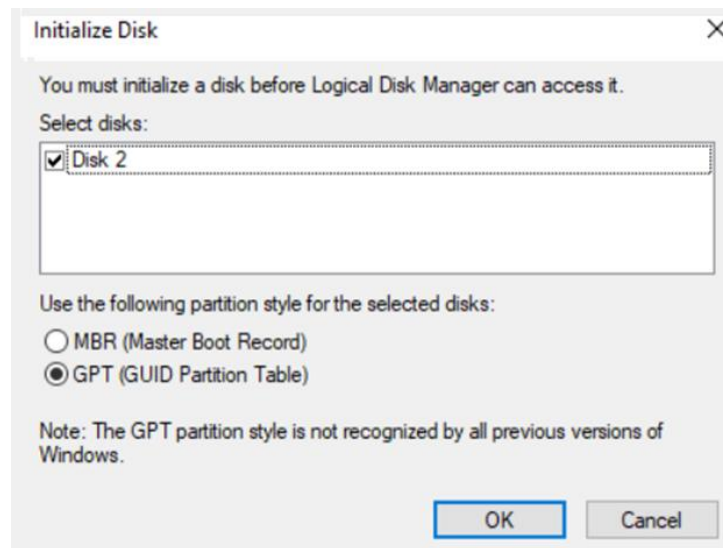


Right Click on Start Menu > Disk Management





Right Click and Initialize Disk



[Choose GPT over MBR](#)

SELECTING A PARTITION STYLE: GPT OR MBR

Windows supports two main disk partition styles: Master Boot Record (MBR) and GUID Partition Table (GPT).

GPT is the standard choice for most modern Windows computers. It supports disks and volumes larger than 2 TB, provides support for more partitions, and stores redundant partition information to improve reliability.

MBR is an older partition style that is mainly used for compatibility with legacy systems and legacy BIOS-based boot configurations. It is limited to approximately 2 TB of usable disk capacity and normally supports up to four primary partitions.

For a Windows boot disk, the partition style must match the firmware mode:

- **UEFI boot normally requires GPT.**
- **Legacy BIOS boot normally uses MBR.**

Additional data disks may use GPT or MBR, but GPT is normally preferred unless a specific compatibility requirement requires MBR.



Feature	MBR	GPT
Typical use	Legacy systems and compatibility requirements	Modern Windows computers and servers
Firmware for Windows boot disk	Legacy BIOS	UEFI
Disk capacity	Approximately 2 TB maximum usable capacity	Supports disks larger than 2 TB
Windows partition limit	Up to four primary partitions, or three primary partitions and one extended partition containing logical drives	Up to 128 partitions on a basic Windows disk
Partition information	Stored primarily in one main partition table	Stores primary and backup partition information
Integrity protection	Does not provide GPT-style CRC protection for the partition table	Uses CRC protection to detect partition-table corruption
Compatibility	Better compatibility with older hardware and Operating Systems	Preferred for modern Windows systems
Recommended use	Only when required by a legacy or compatibility requirement	Default choice for most new Windows deployments

Important distinction:

GPT and MBR are partition styles. NTFS, FAT32, exFAT, and ReFS are file systems.

Selecting GPT does not automatically format a volume as NTFS, and selecting NTFS does not determine whether the disk uses GPT or MBR.

QUICK CHECK – DISK PREPARATION DECISIONS

1. A new 4 TB disk has been added to a modern UEFI-based Windows computer. Which partition style should normally be selected, and why?
2. A technician wants to format the active Windows C: volume from File Explorer while Windows is running. Is this possible?
3. A quick format has completed successfully. Does this prove that all previous data has been securely erased?
4. A disk appears as Not Initialized, but the user says it previously contained important files. What should the technician do first?
5. Is NTFS an alternative to GPT? Explain the difference.



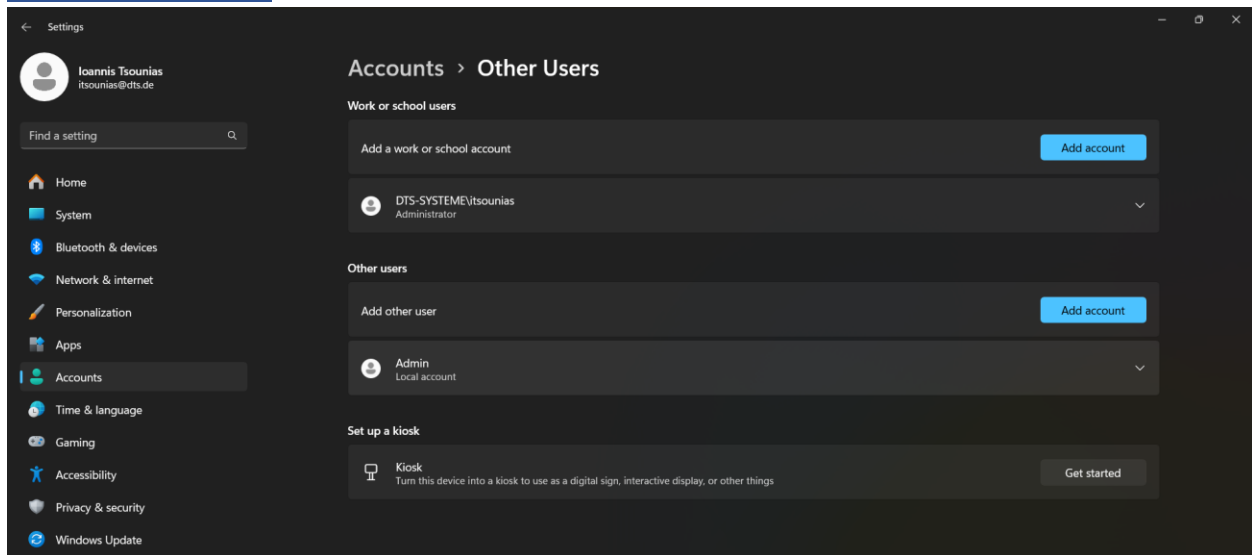
USER ACCOUNT MANAGEMENT IN WINDOWS OS

It is very important to know that when configuring a computer, we should always create two accounts. One account belongs to the Administrator and the other to the User.

The Admin Account has unrestricted access to the entire OS, while on the contrary, for security reason all other Users should have limited access.

Create a New Local User Account

Using the Settings App



Start > Settings > Accounts > Other Users: Add Account

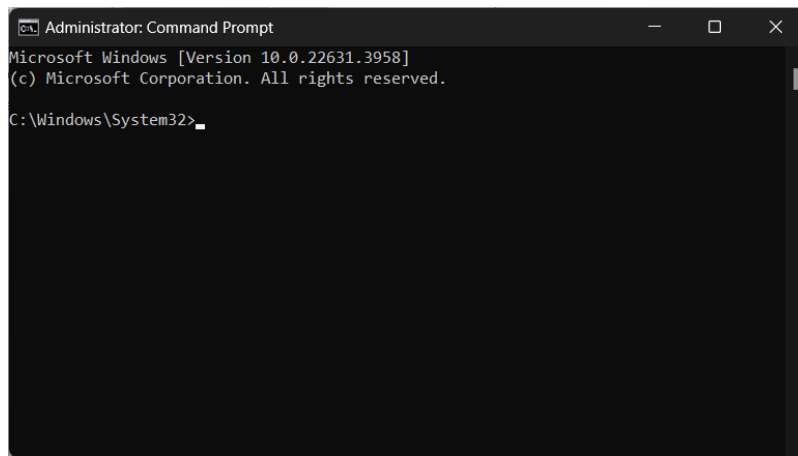
A screenshot of a 'Microsoft account' creation dialog box. The title bar says 'Microsoft account'. The main text reads 'Create a user for this PC'. Below this, there is a paragraph of text: 'If this account is for a child or teenager, consider selecting **Back** and creating a Microsoft account. When younger family members log in with a Microsoft account, they'll have privacy protections focused on their age.' Another paragraph follows: 'If you want to use a password, choose something that will be easy for you to remember but hard for others to guess.' Then, the question 'Who's going to use this PC?' is followed by a text input field labeled 'User name'. Below that, the text 'Make it secure.' is followed by two text input fields labeled 'Enter password' and 'Re-enter password'. A blue 'Next' button is at the bottom right.

Follow the Steps to Create the Account



Using the CMD

It's very interesting and sometimes faster to work with CMD when we want to create a user:



Run the CMD as Admin

The command: net user "username" ""password" /add, (adds a new user).

Then if we need to provide to this new account administrator rights:

The command: net localgroup administrators "username" /add, (admin rights).



User Profiles

Let's begin to clarify what User Profile is in Windows environment.

A Windows user profile is the collection of folders, files, and registry settings that define one user's working environment. When you sign in, Windows loads your profile from C:\Users<your_username> and applies the settings it contains, giving each user on the machine a unique desktop, Documents folder, application settings, and more.

What's inside a user profile folder?

- Desktop, Documents, Downloads, Pictures, Videos: where your personal files live.
- AppData
 - Roaming: application settings and *data that roam with you if you use a network or Microsoft account.*
 - Local: caches and temporary *data tied to this specific PC.*
 - NTUSER.DAT: the user's registry space, storing their personalized registry keys (wallpaper, installed fonts, etc.)

STORAGE

Storage includes the technologies used to retain data, applications, and Operating System files for later use.

Storage may be located inside a computer, attached externally, accessed across a local network, or provided through a cloud service. The appropriate storage solution depends on factors such as capacity, performance, availability, security, compatibility, and cost.

1. Local Storage
 - HDD: (Hard Disk Drive), Classic storage device, big capacity at good prices but not that fast.
 - SSD: (Solid State Drive), Newer storage device, has high speed, more durability, but less capacity at a more expensive price.
 - NVMe SSD (Non-Volatile Memory Express): Is a storage communication protocol, not a separate storage medium. It uses the PCIe interface which gives it direct communication with the CPU, so the speeds are much faster even than an SSD disk.
2. Cloud Storage
 - Public Cloud: Storage services provided by third parties, access via Internet only.
 - Private Cloud: Storage services located within the company's infrastructure.
 - Hybrid Cloud: It is a combination of Public Cloud and Private Cloud; with this implementation the data can be stored in both places.
3. NAS (Network Attached Storage): Device on which we can store data and all legitimate users of the company will have network access.
4. SAN (Storage Area Network): Is a high-speed network that provides network access to storage devices. SANs are typically composed of hosts, switches, storage elements, and storage devices that are interconnected using a variety of technologies, topologies, and protocols.



NAS vs SAN

NAS – Network Attached Storage vs SAN – Storage Area Network	
Connects via LAN	Connects Via High-Speed Networks
Has Its Own IP And Is A Network Node	Uses Switches To Manage Traffic
Operates At The File Level, Providing Access To Files Via Protocols Like SMB/CIFS Or NFS	Operates At The Block Level, Providing Access To Storage Volumes That Can Be Formatted With Different File Systems
Ideal For Small Businesses Or Home Networks	Ideal For Large Enterprises And Data Centers
Easy To Set Up And Manage Through A Web Browser Interface	Provides High Performance And Low Latency, Suitable For Applications Requiring Fast Data Access
Generally, More Affordable Than SAN	Typically, More Expensive Than NAS Due To Complexity And Equipment Requirements

WINDOWS FILE SYSTEM

In every OS, the File System is responsible for storing, organizing, managing, and retrieving data. The File System is used on a storage device such as a hard disk or a CD/DVD or even a USB Stick and is responsible for the proper organization of sectors into directories and files.

The kernel of an OS manages File Storage and the File System. File System management consists of three components: Data, Metadata and File System.

Microsoft File System

The Windows file system is a hierarchical structure that organizes and manages files and directories on a Windows operating system. It provides the framework to store, locate, and access data. The file system is responsible for managing disk space allocation, naming conventions, file retrieval, and security. In early Windows versions **FAT** (File Allocation Table) was the main File System for the OS. But it had some issues, that NTFS (New Technology File System) resolve them. The **NTFS**, improved performance, and security. The most recent Microsoft File System is the **ReFS** (Resilient File System), which is focused on data integrity and fault tolerance.

FAT (File Allocation Table)

The FAT system was used in older Windows versions such as Windows 95 and Windows 98. It is a simple file system, known for its compatibility and simplicity. But, it has limitations in terms of file size support and security. It organizes files in a directory structure and uses a *16-bit* or *32-bit* file allocation table to manage disk space allocation. The FAT system is primarily used for small storage devices and removable media.

FAT (File Allocation Table)	
Pros	Cons
Compatibility with various operating systems	Older File System
Simplicity	Does not support File Encryption
Low Overhead	Does not support Access Control
	Does not support Large Files



NTFS (New Technology File System)

The NTFS is the default file system used in since Windows XP, Windows 7, Win 10 and Win 11. It provides improved performance, security, and stability compared to the FAT system.

NTFS (New Technology File System)	
Pros	Cons
Supports Large Files	Requires more Disk Space for File Storage
Enhanced Security	Non-Compatible with non-Windows Oss
Improved Performance	

ReFS (Resilient File System)

Introduced with Windows Server 2012, **ReFS** is designed to deliver high data integrity, fault tolerance, and scalability for data storage needs in enterprise-level environments. It uses advanced techniques, such as checksums, to ensure data integrity and protection against data corruption. It supports automatic correction of corrupted data and optimized storage usage for large-scale deployments. ReFS is used in server environments for mission-critical applications.

ReFS (Resilient File System)	
Pros	Cons
Enhanced Data Integrity	Lacks in File Compression
Fault Tolerance	Lacks in Encryption
Scalability	